M2201 - Introduction to Graph Theory

TD2

Basic notions and exercises

All the graphs we consider are *simple*, which means they have no loops nor multiple edges. Depending on the cases, they are *directed* or *undirected* (by default, undirected).

1 3-regular graphs.

In a 3-regular graph, all vertices have degree 3 (equivalently, all vertices have 3 neighbors).

- Build such a graph with 4 vertices, 5 vertices, 6 vertices, and 7 vertices.
- What do you think?
- Explain why!

2 4-regular graphs.

Same questions as before, for 4-regular graphs.

3 Graphic sequences.

A sequence of numbers is called *graphic* if we can find a graph whose degrees are these numbers (for example, a triangle graph corresponds to the sequence $2, 2, 2$).

1. Which of the following sequences are graphic?
   - 3, 3, 2, 1, 1
   - 3, 3, 1, 1
   - 3, 3, 2, 2
   - 4, 2, 1, 1, 1, 1
   - 5, 3, 2, 1, 1, 1
   - 5, 4, 3, 1, 1, 1, 1
2. Find two non-isomorphic (= distinct) graphs corresponding to the sequence 3, 2, 2, 2, 1.

4 Distinct degrees.

Back to undirected graphs... Can you find a graph in which all the degrees are different? If yes, draw it. If not why?

5 The six friends.

Given two persons, either they know each other, or they don't (reciprocally). We say that three persons (A, B, and C) know each other if A and B know each other; B and C know each other; and A and C know each other. We say that they don't known each other if *none* of them know each other.

Consider the following declaration: "In a group of five persons, there is *always* three persons who know each other or three persons who do not know each other."

1. Translate this question into a graph question
   Hint: you can represent this social network by a complete graph, whose edges are colored either green (know each other) or red (don't know each other).
2. Is this declaration true?
3. What about 6 persons (instead of 5)?

6 Graphic sequences for directed graphs.

We now apply the same idea for *directed* graphs. Since a vertex has both an *in-degree* and an *out-degree*, we must now consider *couples* of numbers (in-degree, out-degree). Which of the following sequences are graphic?

- (0,1), (1,1), (1,1), (1,1), (1,0)
- (1,1), (1,1), (1,1), (1,1), (1,1)
- (0,2), (1,1), (1,1), (1,1)
- (0,2), (1,1), (1,1), (2,0)
- (1,2), (1,2), (2,1), (2,1)
- (1,2), (1,2), (2,1), (2,2), (1,1)